

SE4050 – Deep Learning

Week 7 – Recurrent Neural Networks and Mid-Term Recap

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Mid term recap

- Lecture 1
 - What is deep learning ?
 - AI Vs ML Vs DL
 - Connection between brain neural networks and artificial neural networks
 - Three generations of Artificial Neurons
 - Multilayer Perceptron
 - Data amount vs size of the DNN
 - Challenges of DL
 - When to use DL when not to use DL

- Lecture 2
 - Parts of a biological neuron
 - Activation functions, weights and biases
 - Linearly and non-linearly separable data
 - Training sequence of a DNN
 - Forward propagations
 - Backpropagation learning
 - Loss function
 - Derivatives
 - Chain rule

- Lecture 3
 - Optimization algorithms
 - GD
 - SGD
 - MGD
 - Adam
 - Regularization
 - Bias
 - Variance
 - L1 and L2
 - Basic Machine Learning
 - Split test and cross-validation
 - Supervised, Unsupervised, Semi-supervised, Reinforcement
 - Evaluation Metrics: Confusion matrix, Accuracy, precision, recall, F1-score, ROC-AUC, for classification problems, and RMSE, MAE, R^2 for regression.
 - Data preprocessing – Normalization, Scaling, Encoding
 - Overfitting, Underfitting, Generalization

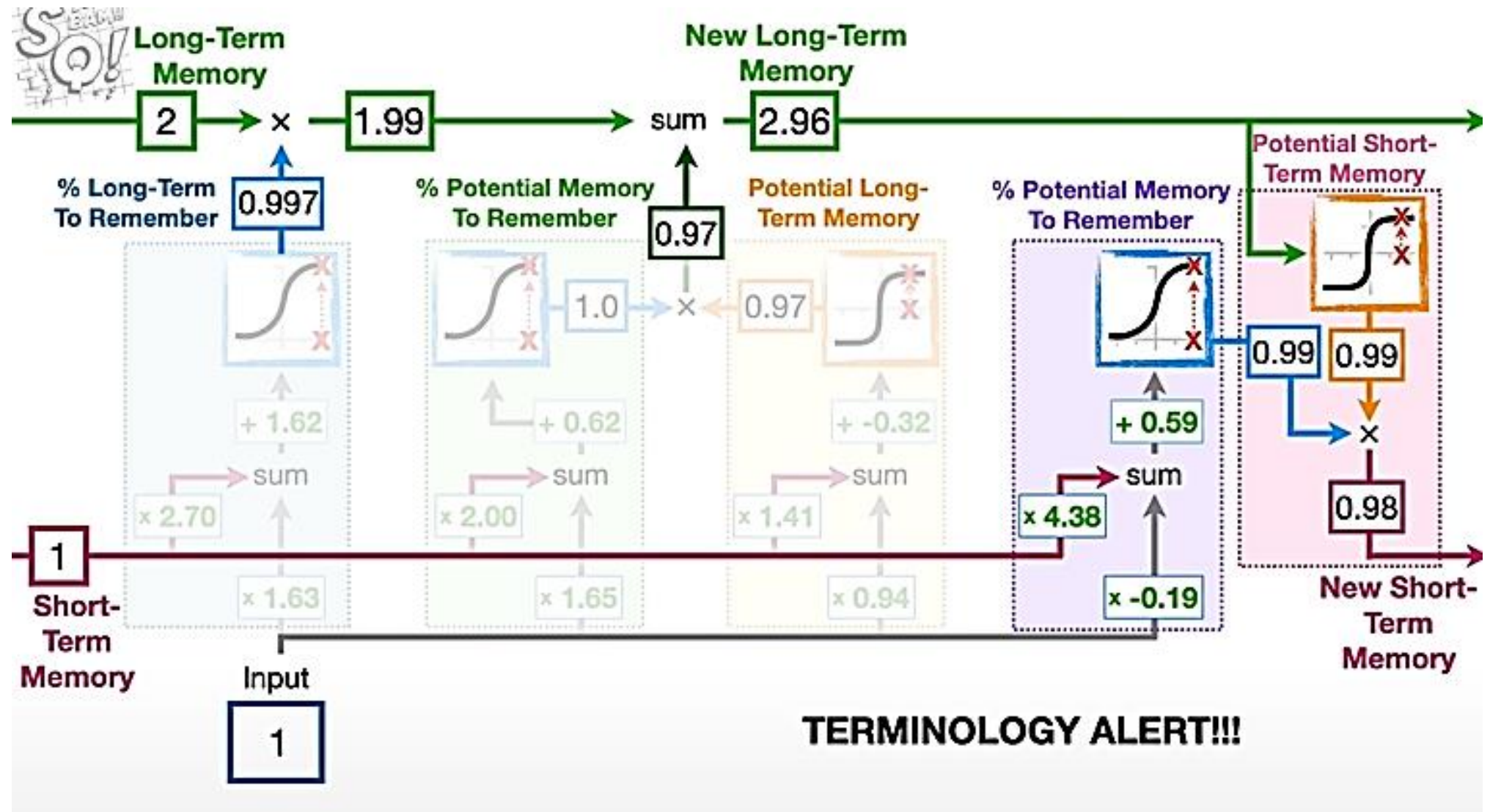
- Lecture 4
 - Scalar, Vector, Matrix, Tensor
 - Matrices and neural networks
 - CNNs
 - Convolution (1D, 2D, 3D)
 - Kernels and feature maps
 - Pooling (Global vs non-global)
 - Stride
 - Size
 - Padding
 - Softmax function
 - Categorical cross entropy
 - How to prevent over-fitting

- Lecture 5
 - How to select better optimization algorithms
 - Usage vs Loss functions
 - Transfer Learning steps
 - Data augmentation
 - Traditional
 - Deep learning

- Lecture 6
 - Types of data
 - Dimensions
 - Spatiotemporal data
 - Sequential vs non-sequential data
 - Processing sequences
 - Recurrent neural networks
 - Difference between RNN and normal DNN
 - Memory
 - Unfolding/unrolling
 - Weight and bias sharing
 - Forwards propagation and backpropagation in time
 - Vanishing and exploding gradient
 - LSTMs
 - Unit structure and functions
 - Forget gate
 - Input gate
 - Output gate
 - Using RNNs and LSTMs

Why share weight in an RNN?

LSTM recap (from StatQuest)



Vanishing Gradient Solution: LSTM's cell state and gating mechanisms (forget gate, input gate, output gate) allow the network to maintain long-term dependencies without the gradients vanishing.

Exploding Gradient Solution: LSTM networks can use techniques like gradient clipping and leverage the gating mechanisms to regulate the flow of gradients, preventing them from becoming too large.

By managing both vanishing and exploding gradients, LSTMs are highly effective for learning long-term dependencies and have become a standard architecture in sequential data tasks such as natural language processing, time series prediction, and more.